

Mathematical Properties of the Affine-DDPNM

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1 Preliminaries and notation

We adopt the domain partition, finite-element discretisation, and notation introduced in the accompanying methodology document. The pore domain $\Omega \subset \mathbb{R}^3$ is partitioned into N non-overlapping subdomains $\{\Omega_i\}_{i=1}^N$ with m internal interfaces $\{\gamma_k\}_{k=1}^m$. Subdomain Ω_i has m_i ports, of which m_i^u are internal (unknown-traction) and m_i^k are boundary (known-traction).

Interface traction basis. On each interface γ , we introduce a local orthonormal frame $\{\mathbf{n}_\gamma, \mathbf{t}_{\gamma,1}, \mathbf{t}_{\gamma,2}\}$ with origin at the interface centroid $\mathbf{c}_\gamma$, and dimensionless in-plane coordinates $s(\mathbf{x}), t(\mathbf{x}) = O(1)$ on γ . The *mode index set* is

$$\mathcal{M} = \{0, s, t\} \times \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}, \quad (1)$$

with scalar shape functions $\varphi_0 = 1$, $\varphi_s = s$, $\varphi_t = t$ and direction vectors $\mathbf{d}_\mathbf{n} = \mathbf{n}_\gamma$, $\mathbf{d}_{\mathbf{t}_1} = \mathbf{t}_{\gamma,1}$, $\mathbf{d}_{\mathbf{t}_2} = \mathbf{t}_{\gamma,2}$. The traction on γ is expanded as

$$\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n}_\gamma = \sum_{(\alpha, \beta) \in \mathcal{M}} \lambda_{\gamma, \alpha, \beta} \varphi_\alpha(s, t) \mathbf{d}_\beta. \quad (2)$$

For a subdomain Ω_i , let $n_i = 9m_i^u + m_i^k$ be the total number of local traction unknowns (9 per internal interface, 1 per boundary port). Collect them in the vector $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$. The restriction maps R_i^u, R_i^k and the global unknown vector $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$ are defined analogously to the classic case, with the block size 9×9 per interface instead of 1×1 .

Unit-response loads. For each mode $(\alpha, \beta) \in \mathcal{M}$ on an internal interface $\gamma_{i,r}$ of subdomain Ω_i , the Neumann load vector $\mathbf{f}_{i,r}^{(\alpha, \beta)} \in \mathbb{R}^{n_{i,u}}$ has entries

$$[\mathbf{f}_{i,r}^{(\alpha, \beta)}]_j = - \int_{\gamma_{i,r}} \varphi_\alpha(s, t) \mathbf{d}_\beta \cdot \boldsymbol{\phi}_j^{(i)} dS, \quad (3)$$

where $\{\boldsymbol{\phi}_j^{(i)}\}$ is the local velocity basis. For boundary ports only the uniform-normal mode $(\alpha, \beta) = (0, \mathbf{n})$ is used, with a *prescribed* coefficient equal to the inlet/outlet pressure.

Local Stokes system. For each unit load, the discrete Taylor–Hood system on Ω_i reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{i,r}^{(\alpha, \beta)} \\ \mathbf{X}_{i,r}^{(\alpha, \beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,r}^{(\alpha, \beta)} \\ \mathbf{0} \end{bmatrix}, \quad (4)$$

where $A_i \in \mathbb{R}^{n_{i,u} \times n_{i,u}}$ is the velocity-stiffness matrix (SPD) and $B_i \in \mathbb{R}^{n_{i,p} \times n_{i,u}}$ is the divergence matrix. The pair $(V_{i,h}, Q_{i,h})$ is the inf–sup stable Taylor–Hood $[P_2]^3$ – P_1 element.

For a boundary port $\gamma_{i,r} \in \mathcal{K}_i$, the single $(\alpha, \beta) = (0, \mathbf{n})$ load is used.

Local DtN operator. The local Dirichlet-to-Neumann (traction-to-flux) map $\mathbf{G}_i \in \mathbb{R}^{n_i \times n_i}$ has entries

$$\mathbf{G}_i[(\gamma, \alpha, \beta), (\gamma', \alpha', \beta')] = -[\mathbf{f}_i^{(\alpha, \beta)}(\gamma)]^\top \mathbf{U}_i^{(\alpha', \beta')}(\gamma'), \quad (5)$$

where (γ, α, β) ranges over the n_i local modes. For boundary ports, only the block $(\gamma, 0, \mathbf{n})$ is stored. $\mathbf{G}_i$ is block-partitioned into $\mathbf{G}_i^{uu}$ (unknown–unknown) and $\mathbf{G}_i^{uk}$ (unknown–known) as in the classic case.

Global Schur complement. Enforcing flux continuity on every internal interface (constant-mode normal component only) yields the global system

$$\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}, \quad \mathbf{S} = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k, \quad (6)$$

where $\boldsymbol{\lambda}^k \in \mathbb{R}^{m^k}$ collects the prescribed inlet/outlet tractions and $\mathbf{S} \in \mathbb{R}^{9m \times 9m}$.

2 Mathematical properties

We now establish the solvability, symmetry, positive-definiteness, and mass-conservation properties of the Affine-DDPNM formulation. The overall structure parallels the classic DD-PNM analysis [1], with careful treatment of the nine-mode interface space.

Lemma 2.1 (Local well-posedness). *For every interface-traction vector $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$, the discrete Stokes subproblem in each Ω_i admits a unique solution $(\mathbf{U}_i, \mathbf{X}_i)$.*

Proof. The saddle-point matrix $K_i = \begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix}$ is invertible because (i) A_i is SPD and (ii) the Taylor–Hood pair satisfies the discrete inf–sup condition on the body-fitted tetrahedral mesh of Ω_i . The right-hand side of (4) is a linear combination of the primitive load vectors $\mathbf{f}_{i,r}^{(\alpha,\beta)}$ with coefficients $\boldsymbol{\lambda}_i$; hence a unique solution exists for any $\boldsymbol{\lambda}_i$. $\square$

Lemma 2.2 (Discrete incompressibility of the unit responses). *For each internal interface $\gamma_{i,r}$ and each mode $(\alpha, \beta) \in \mathcal{M}$, the velocity response $\mathbf{U}_{i,r}^{(\alpha,\beta)}$ is discretely divergence-free:*

$$B_i \mathbf{U}_{i,r}^{(\alpha,\beta)} = \mathbf{0}. \quad (7)$$

Proof. Equation (7) follows directly from the second block row of the local system (4). For completeness, we also provide the explicit expression of the response. From the first block row,

$$\mathbf{U}_{i,r}^{(\alpha,\beta)} = A_i^{-1} \left(\mathbf{f}_{i,r}^{(\alpha,\beta)} - B_i^\top \mathbf{X}_{i,r}^{(\alpha,\beta)} \right).$$

Imposing $B_i \mathbf{U}_{i,r}^{(\alpha,\beta)} = \mathbf{0}$ yields

$$\mathbf{X}_{i,r}^{(\alpha,\beta)} = M_i^{-1} B_i A_i^{-1} \mathbf{f}_{i,r}^{(\alpha,\beta)}, \quad M_i := B_i A_i^{-1} B_i^\top \text{ (SPD)}.$$

Substituting back,

$$\mathbf{U}_{i,r}^{(\alpha,\beta)} = \underbrace{(A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1})}_{=: H_i} \mathbf{f}_{i,r}^{(\alpha,\beta)}, \quad (8)$$

where H_i is the velocity–velocity block of K_i^{-1} . Since $B_i H_i = 0$ (the operator H_i maps any load vector into $\ker B_i$), we have $B_i \mathbf{U}_{i,r}^{(\alpha,\beta)} = \mathbf{0}$. $\square$

Lemma 2.3 (Symmetry and energy identity for the local DtN map). *Let $\mathbf{G}_i$ be defined by (5). Then $\mathbf{G}_i$ is symmetric and, for any local traction vector $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$,*

$$\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = -\mathbf{U}_i^\top A_i \mathbf{U}_i \leq 0, \quad (9)$$

where $\mathbf{U}_i = \sum_{\gamma,\alpha,\beta} \lambda_{i,\gamma,\alpha,\beta} \mathbf{U}_i^{(\alpha,\beta)}(\gamma)$ is the velocity field induced by $\boldsymbol{\lambda}_i$. In particular, $\mathbf{G}_i$ is symmetric negative semi-definite.

Proof. Denote the two index pairs by $\mu = (\gamma, \alpha, \beta)$ and $\mu' = (\gamma', \alpha', \beta')$. By Lemma 2.2, $B_i \mathbf{U}_i^{(\mu')} = \mathbf{0}$. Using the explicit form (8) and the symmetry of A_i ,

$$\begin{aligned} [\mathbf{f}_i^{(\mu)}]^\top \mathbf{U}_i^{(\mu')} &= [\mathbf{f}_i^{(\mu)}]^\top H_i \mathbf{f}_i^{(\mu')} \\ &= [H_i \mathbf{f}_i^{(\mu)}]^\top A_i [H_i \mathbf{f}_i^{(\mu')}] \\ &= [\mathbf{U}_i^{(\mu)}]^\top A_i \mathbf{U}_i^{(\mu')}, \end{aligned}$$

where we used $H_i^\top A_i H_i = H_i$ (a standard property of the Schur complement on the velocity space).

The final expression is symmetric in (μ, μ') , hence $\mathbf{G}_i$ is symmetric. For the energy identity, expand

$$\begin{aligned} \boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i &= - \sum_{\mu, \mu'} \lambda_{i, \mu} \lambda_{i, \mu'} [\mathbf{f}_i^{(\mu)}]^\top \mathbf{U}_i^{(\mu')} \\ &= - \left(\sum_{\mu} \lambda_{i, \mu} \mathbf{U}_i^{(\mu)} \right)^\top A_i \left(\sum_{\mu'} \lambda_{i, \mu'} \mathbf{U}_i^{(\mu')} \right) \\ &= - \mathbf{U}_i^\top A_i \mathbf{U}_i \leq 0, \end{aligned}$$

where the non-positivity follows because A_i is SPD. $\square$

Lemma 2.4 (Kernel of the local DtN map). *Let $\mathbf{1}_{(0, \mathbf{n})} \in \mathbb{R}^{n_i}$ be the vector with entry 1 at every internal-interface uniform-normal mode $(\gamma, 0, \mathbf{n})$ and entry 0 elsewhere; and with entry 1 at every boundary-port (single) mode. Then*

(i) $\mathbf{G}_i \mathbf{1}_{(0, \mathbf{n})} = \mathbf{0}$.

(ii) $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = 0$ if and only if $\boldsymbol{\lambda}_i \in \text{span}\{\mathbf{1}_{(0, \mathbf{n})}\}$, i.e. all non-zero components of $\boldsymbol{\lambda}_i$ are confined to the uniform-normal modes and share a common coefficient.

Moreover, the kernel of $\mathbf{G}_i$ is exactly one-dimensional.

Proof. (i) Set all internal uniform-normal coefficients to the same value p_{const} and all other coefficients to zero. The right-hand side of the local Stokes system (4) becomes

$$\sum_{\gamma} p_{\text{const}} \mathbf{f}_i^{(\gamma, 0, \mathbf{n})} = p_{\text{const}} \sum_{\gamma} \left(- \int_{\gamma_{i, \gamma}} \boldsymbol{\phi}_j^{(i)} \cdot \mathbf{n}_{i, \gamma} \, dS \right).$$

By the divergence theorem (noting that each $\boldsymbol{\phi}_j^{(i)}$ vanishes on solid walls),

$$\sum_{\gamma} \mathbf{f}_i^{(\gamma, 0, \mathbf{n})} = - \int_{\partial \Omega_i} \boldsymbol{\phi}_j^{(i)} \cdot \mathbf{n} \, dS = - \int_{\Omega_i} \nabla \cdot \boldsymbol{\phi}_j^{(i)} \, d\mathbf{x}.$$

Let $p_{i, h} \equiv p_{\text{const}}$ be the constant pressure function with coefficient vector $\mathbf{X}_{i, \text{const}}$. By definition of B_i ,

$$[B_i^\top \mathbf{X}_{i, \text{const}}]_j = - \int_{\Omega_i} p_{\text{const}} \nabla \cdot \boldsymbol{\phi}_j^{(i)} \, d\mathbf{x} = \sum_{\gamma} p_{\text{const}} \mathbf{f}_i^{(\gamma, 0, \mathbf{n})}.$$

Hence $(\mathbf{U}_i, \mathbf{X}_i) = (\mathbf{0}, \mathbf{X}_{i, \text{const}})$ solves (4) for this traction vector. By uniqueness (Lemma 2.1), $\mathbf{U}_i = \mathbf{0}$. Applying the energy identity (9) gives $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = 0$, and since $\mathbf{G}_i$ is symmetric negative semi-definite, this implies $\mathbf{G}_i \boldsymbol{\lambda}_i = \mathbf{0}$.

(ii) $(\Leftarrow)$ is proved in (i).

$(\Rightarrow)$ Assume $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = 0$. By the energy identity (9), $\mathbf{U}_i^\top A_i \mathbf{U}_i = 0$, so $\mathbf{U}_i = \mathbf{0}$ because A_i is SPD. The first block row of (4) then gives

$$B_i^\top \mathbf{X}_i = \sum_{\gamma, \alpha, \beta} \lambda_{i, \gamma, \alpha, \beta} \mathbf{f}_i^{(\gamma, \alpha, \beta)}. \quad (10)$$

We must show that the right-hand side belongs to $\text{Range}(B_i^\top)$ only when all non-uniform-normal coefficients vanish and the uniform-normal coefficients share a common value.

Let F_i be the matrix whose columns are all primitive load vectors $\mathbf{f}_i^{(\gamma, \alpha, \beta)}$ for internal interfaces (plus the single column per boundary port). Equation (10) states $F_i \boldsymbol{\lambda}_i \in \text{Range}(B_i^\top)$.

Consider any $\mathbf{U} \in \ker B_i$ (a discretely divergence-free velocity) and let $\mathbf{v}_h$ be the corresponding FE function. Taking the inner product with the coefficient vector V of $\mathbf{v}_h$,

$$V^\top F_i \boldsymbol{\lambda}_i = \sum_{\gamma, \alpha, \beta} \lambda_{i, \gamma, \alpha, \beta} \left(- \int_{\gamma_{i, \gamma}} \varphi_\alpha(s, t) \mathbf{d}_\beta \cdot \mathbf{v}_h \, dS \right) = 0 \quad \forall V \in \ker B_i.$$

Define the flux map $\Phi : \ker B_i \rightarrow \mathbb{R}^{n_i}$ by

$$[\Phi(V)]_{\gamma, \alpha, \beta} = \int_{\gamma_{i, \gamma}} \varphi_\alpha(s, t) \mathbf{d}_\beta \cdot \mathbf{v}_h \, dS.$$

The condition above is $\boldsymbol{\lambda}_i \perp \text{Range}(\Phi)$.

We now characterise $\text{Range}(\Phi)$. For the constant-normal mode $(\alpha, \beta) = (0, \mathbf{n})$, the flux components satisfy the zero-sum constraint $\sum_\gamma [\Phi(V)]_{\gamma, 0, \mathbf{n}} = 0$ (by the divergence theorem and $V \in \ker B_i$). Moreover, by an argument parallel to Lemma A.2 of [1], the map Φ restricted to the constant-normal components is surjective onto the zero-sum subspace of $\mathbb{R}^{m_i^u + m_i^k}$.

For all other modes $(\alpha, \beta) \neq (0, \mathbf{n})$, the scalar shape φ_α has zero mean on $\gamma_{i, \gamma}$ (since $\int_\gamma s \, dS = \int_\gamma t \, dS = 0$ by construction of the centroid-centred coordinates), and the tangential directions $\mathbf{d}_{t_1}, \mathbf{d}_{t_2}$ provide independent channels. An adaptation of Lemma A.2 of [1] to the richer flux space shows that the non-constant-normal mode components have full range (no zero-sum constraint). Consequently, orthogonality to $\text{Range}(\Phi)$ forces all non- $(0, \mathbf{n})$ coefficients to vanish and the $(0, \mathbf{n})$ coefficients to be equal.

Therefore $\boldsymbol{\lambda}_i \in \text{span}\{\mathbf{1}_{(0, \mathbf{n})}\}$, completing the converse direction.

The kernel is one-dimensional because any vector in the kernel is a scalar multiple of $\mathbf{1}_{(0, \mathbf{n})}$, and $\mathbf{1}_{(0, \mathbf{n})} \neq \mathbf{0}$. $\square$

Remark 2.5. Lemma 2.4 has a clear physical interpretation: the only way to obtain zero velocity in a pore subdomain (which has at least one no-slip solid wall, fixing rigid-body modes) is to apply a uniform constant pressure on all its boundaries. Any spatial variation of the normal traction (the $\mathbf{sn}$, $\mathbf{tn}$ modes) or any tangential traction component drives a non-trivial Stokes flow. The kernel therefore remains one-dimensional despite the nine-fold enrichment of the interface space.

Theorem 2.6 (Solvability and mass conservation of Affine-DDPNM). *Let $\mathbf{S} \in \mathbb{R}^{9m \times 9m}$ be the global Schur complement defined in (6) and let $\mathbf{F}$ be the corresponding forcing vector. Then:*

- (i) **Positive definiteness.** $\mathbf{S}$ is symmetric positive-definite (SPD).
- (ii) **Global solvability.** The linear system $\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}$ has a unique solution $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$.
- (iii) **Mass conservation.** The reconstructed velocity field $\mathbf{u}_h^{DD} = \sum_i \mathbf{u}_{i, h}$ satisfies both local and global discrete mass balances:

$$\sum_{r=1}^{m_i} Q_{i, r}^{(0, \mathbf{n})} = 0 \quad (\text{local, each } \Omega_i), \quad \int_{\partial \Omega} \mathbf{u}_h^{DD} \cdot \mathbf{n} \, dS = 0 \quad (\text{global}). \quad (11)$$

Proof. (i) **Symmetry and positive definiteness.** Symmetry of $\mathbf{S}$ follows from the symmetry of each $\mathbf{G}_i^{uu}$ (Lemma 2.3) and the Boolean nature of the restriction matrices R_i^u .

For any $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$, let $\boldsymbol{\lambda}_i^u = R_i^u \boldsymbol{\lambda}$ be the local unknown-traction vector on Ω_i (with zero entries for boundary ports). By the energy identity (9),

$$\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = - \sum_{i=1}^N (\boldsymbol{\lambda}_i^u)^\top \mathbf{G}_i^{uu} \boldsymbol{\lambda}_i^u = \sum_{i=1}^N \mathbf{U}_i^\top A_i \mathbf{U}_i \geq 0.$$

If $\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = 0$, then $\mathbf{U}_i = \mathbf{0}$ for every i . By Lemma 2.4(ii), each $\boldsymbol{\lambda}_i^u$ must belong to $\text{span}\{\mathbf{1}_{(0,\mathbf{n})}\}$ on the internal interfaces of Ω_i .

Global consistency across adjacent pores forces all uniform-normal coefficients to share the same value $c \in \mathbb{R}$ on every internal interface. However, the boundary ports of each pore carry *prescribed* (not unknown) tractions. The only constant value compatible with $\boldsymbol{\lambda}_i^u$ having zero entries on boundary ports (since $\boldsymbol{\lambda}_i^u$ is formed by restriction of the purely internal vector $\boldsymbol{\lambda}$) is $c = 0$. Hence $\boldsymbol{\lambda} = \mathbf{0}$, and $\mathbf{S}$ is SPD.

(ii) **Global solvability.** Immediate from (i): an SPD matrix is invertible.

(iii) **Mass conservation.** For each subdomain Ω_i , choose the constant pressure test function $q_h \equiv 1 \in Q_{i,h}$ in the discrete incompressibility constraint $B_i \mathbf{U}_i = \mathbf{0}$. By the divergence theorem and the no-slip condition on solid walls,

$$0 = \int_{\Omega_i} \nabla \cdot \mathbf{u}_{i,h} \, d\mathbf{x} = \sum_{r=1}^{m_i} \int_{\gamma_{i,r}} \mathbf{u}_{i,h} \cdot \mathbf{n}_{i,r} \, dS = \sum_{r=1}^{m_i} Q_{i,r}^{(0,\mathbf{n})},$$

which establishes the local mass balance.

The global Schur assembly (6) enforces $Q_{i,r}^{(0,\mathbf{n})} + Q_{j,r}^{(0,\mathbf{n})} = 0$ on every internal interface shared by pores i and j . Summing the local balances over all pores, the internal flux pairs cancel pairwise and only the inlet/outlet contributions remain, yielding $\int_{\partial\Omega} \mathbf{u}_h^{\text{DD}} \cdot \mathbf{n} \, dS = 0$. $\square$

3 Error estimate as a Galerkin projection

Following Appendix B of [1], we formulate the classical non-overlapping domain-decomposition (C-DD) discrete Stokes system and characterise the Affine-DDPNM as its Galerkin restriction onto the nine-mode affine interface space.

C-DD formulation (recap). Let $\boldsymbol{\lambda}^{\text{FE}} \in \mathbb{R}^{m_\Gamma}$ collect the nodal normal and tangential traction degrees of freedom on all internal interfaces, and let $\hat{\mathbf{S}}$ be the C-DD Schur complement (the Steklov–Poincaré operator of the full FE discretisation). The C-DD system is $\hat{\mathbf{S}} \boldsymbol{\lambda}^{\text{FE}} = \hat{\mathbf{F}}$, with $\hat{\mathbf{S}}$ SPD and the energy identity $\|\boldsymbol{\lambda} - \tilde{\boldsymbol{\lambda}}\|_{\hat{\mathbf{S}}}^2 = \sum_i \|\mathbf{U}_i(\boldsymbol{\lambda}) - \mathbf{U}_i(\tilde{\boldsymbol{\lambda}})\|_{A_i}^2$ holding for any two traction vectors.

Affine restriction operator. For each interface γ_k , define the local restriction $P_k^{\text{aff}} \in \mathbb{R}^{m_{\gamma_k} \times 9}$ that maps the nine affine coefficients $\boldsymbol{\lambda}_k \in \mathbb{R}^9$ to the nodal traction vector on γ_k by evaluating the affine Ansatz (2) at each mesh node $\mathbf{x}_\ell$ on γ_k :

$$[P_k^{\text{aff}} \boldsymbol{\lambda}_k]_\ell = \sum_{(\alpha,\beta) \in \mathcal{M}} \lambda_{k,\alpha,\beta} \varphi_\alpha(s(\mathbf{x}_\ell), t(\mathbf{x}_\ell)) \mathbf{d}_\beta \cdot (\text{FE basis at } \mathbf{x}_\ell). \quad (12)$$

The global restriction operator is the block-diagonal assembly $\mathcal{P}_{\text{aff}} = \bigoplus_k P_k^{\text{aff}} : \mathbb{R}^{9m} \rightarrow \mathbb{R}^{m_\Gamma}$.

For the classic DD-PNM, the restriction is $\mathcal{P}_{\text{P0}} : \mathbb{R}^m \rightarrow \mathbb{R}^{m_\Gamma}$, with one column per interface: the L^2 -projection of the constant function onto the nodal FE trace basis, in the normal direction only. Clearly $\text{Range}(\mathcal{P}_{\text{P0}}) \subset \text{Range}(\mathcal{P}_{\text{aff}})$, since the constant normal mode is one of the nine affine modes.

Galerkin projection. Substituting $\boldsymbol{\lambda}^{\text{FE}} = \mathcal{P}_{\text{aff}} \boldsymbol{\lambda}$ into the C-DD system and applying $\mathcal{P}_{\text{aff}}^\top$ yields the Affine-DDPNM system (6) with

$$\mathbf{S} = \mathcal{P}_{\text{aff}}^\top \hat{\mathbf{S}} \mathcal{P}_{\text{aff}}, \quad \mathbf{F} = \mathcal{P}_{\text{aff}}^\top \hat{\mathbf{F}}.$$

Theorem 3.1 (Error estimate for Affine-DDPNM). *Let $\boldsymbol{\lambda}^{\text{FE}}$ be the C-DD interface traction ($\hat{\mathbf{S}} \boldsymbol{\lambda}^{\text{FE}} = \hat{\mathbf{F}}$). Let $\boldsymbol{\lambda}^{\text{aff}}$ be the Affine-DDPNM solution ($\mathbf{S} \boldsymbol{\lambda}^{\text{aff}} = \mathbf{F}$) and set $\boldsymbol{\lambda}_\Gamma^{\text{aff}} = \mathcal{P}_{\text{aff}} \boldsymbol{\lambda}^{\text{aff}}$. Then:*

$$\|\boldsymbol{\lambda}^{\text{FE}} - \boldsymbol{\lambda}_\Gamma^{\text{aff}}\|_{\hat{\mathbf{S}}} = \min_{\boldsymbol{\mu} \in \mathbb{R}^{9m}} \|\boldsymbol{\lambda}^{\text{FE}} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\hat{\mathbf{S}}}, \quad (13)$$

and the velocity energy error satisfies

$$\sum_{i=1}^N \|\mathbf{U}_i^{FE} - \mathbf{U}_i^{aff}\|_{A_i}^2 = \|\boldsymbol{\lambda}^{FE} - \boldsymbol{\lambda}_\Gamma^{aff}\|_{\hat{\mathbf{S}}}^2. \quad (14)$$

Proof. The Galerkin orthogonality $\mathcal{P}_{\text{aff}}^\top \hat{\mathbf{S}}(\boldsymbol{\lambda}_\Gamma^{aff} - \boldsymbol{\lambda}^{FE}) = \mathbf{0}$ follows from subtracting the C-DD and Affine-DDPNM systems. Hence the error $\mathbf{e} := \boldsymbol{\lambda}^{FE} - \boldsymbol{\lambda}_\Gamma^{aff}$ is $\hat{\mathbf{S}}$ -orthogonal to $\text{Range}(\mathcal{P}_{\text{aff}})$. For any $\boldsymbol{\mu} \in \mathbb{R}^{9m}$, applying the Pythagoras identity in the $\hat{\mathbf{S}}$ -inner product gives

$$\|\boldsymbol{\lambda}^{FE} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\hat{\mathbf{S}}}^2 = \|\mathbf{e}\|_{\hat{\mathbf{S}}}^2 + \|\boldsymbol{\lambda}_\Gamma^{aff} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\hat{\mathbf{S}}}^2 \geq \|\mathbf{e}\|_{\hat{\mathbf{S}}}^2,$$

with equality when $\mathcal{P}_{\text{aff}} \boldsymbol{\mu} = \boldsymbol{\lambda}_\Gamma^{aff}$, i.e. when $\boldsymbol{\mu} = \boldsymbol{\lambda}_\Gamma^{aff}$. This proves (13). Equation (14) follows from the C-DD energy identity applied to $\boldsymbol{\lambda}^{FE}$ and $\boldsymbol{\lambda}_\Gamma^{aff}$. $\square$

Remark 3.2. Theorem 3.1 shows that the Affine-DDPNM velocity error equals the minimal distance, in the C-DD Schur energy norm, between the exact FE interface traction and any traction field representable by the nine-mode-per-interface affine Ansatz. Because $\text{Range}(\mathcal{P}_{P0}) \subset \text{Range}(\mathcal{P}_{\text{aff}})$, the minimal distance for the affine space is never larger than that for the classic constant-normal space. This provides a rigorous justification for the 73–90% error reduction observed in the numerical experiments.

Remark 3.3. The projection error in (13) is determined by how well the true FE interface traction can be represented by an affine function in the local frame. Mesh refinement reduces the FE discretisation error of the reference solution but does not shrink the projection error unless the interface space is further enriched (e.g. by adding quadratic shape functions $\{s^2, st, t^2\}$). This explains the numerical observation that the Affine-DDPNM errors are essentially mesh-independent within the tested range (4.3K–45K cells).

References

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